

GEOGRAPHY

INDIAN GEOGRAPHY



INDIAN SOILS

MODULE - 4

SOIL TYPES

- ALLUVIAL SOILS

CLASSIFICATION OF SOILS

- 1 India has varied relief features, landforms, climatic realms and vegetation types.
- 2 These have contributed in the development of various types of soils in India.
- 3 In ancient times, soils used to be classified into two main groups – Urvara and Usara, which were fertile and sterile, respectively.
- 4 In the 16th century A.D., soils were classified on the basis of their inherent characteristics and external features such as texture, colour, slope of land and moisture content in the soil.
- 5 Based on texture, main soil types were identified as sandy, clayey, silty & loam etc.
- 6 On the basis of colour, they are classified into red, yellow, black, etc.
- 7 Since Independence, scientific surveys of soils have been conducted by various agencies.
- 8 Soil Survey of India, established in 1956, made comprehensive studies of soils in selected areas like in the Damodar Valley.
- 9 The National Bureau of Soil Survey and the Land Use Planning Institute under the control of the Indian Council of Agricultural Research (ICAR) did a lot of studies on Indian soils.

CLASSIFICATION OF SOILS

10 In their effort to study soil and to make it comparable at the international level, the ICAR has classified the Indian soils on the basis of their nature and character as per the United States Department of Agriculture (USDA) Soil Taxonomy.



11 On the basis of Genesis, colour, composition and location, the soils of India have been classified into:

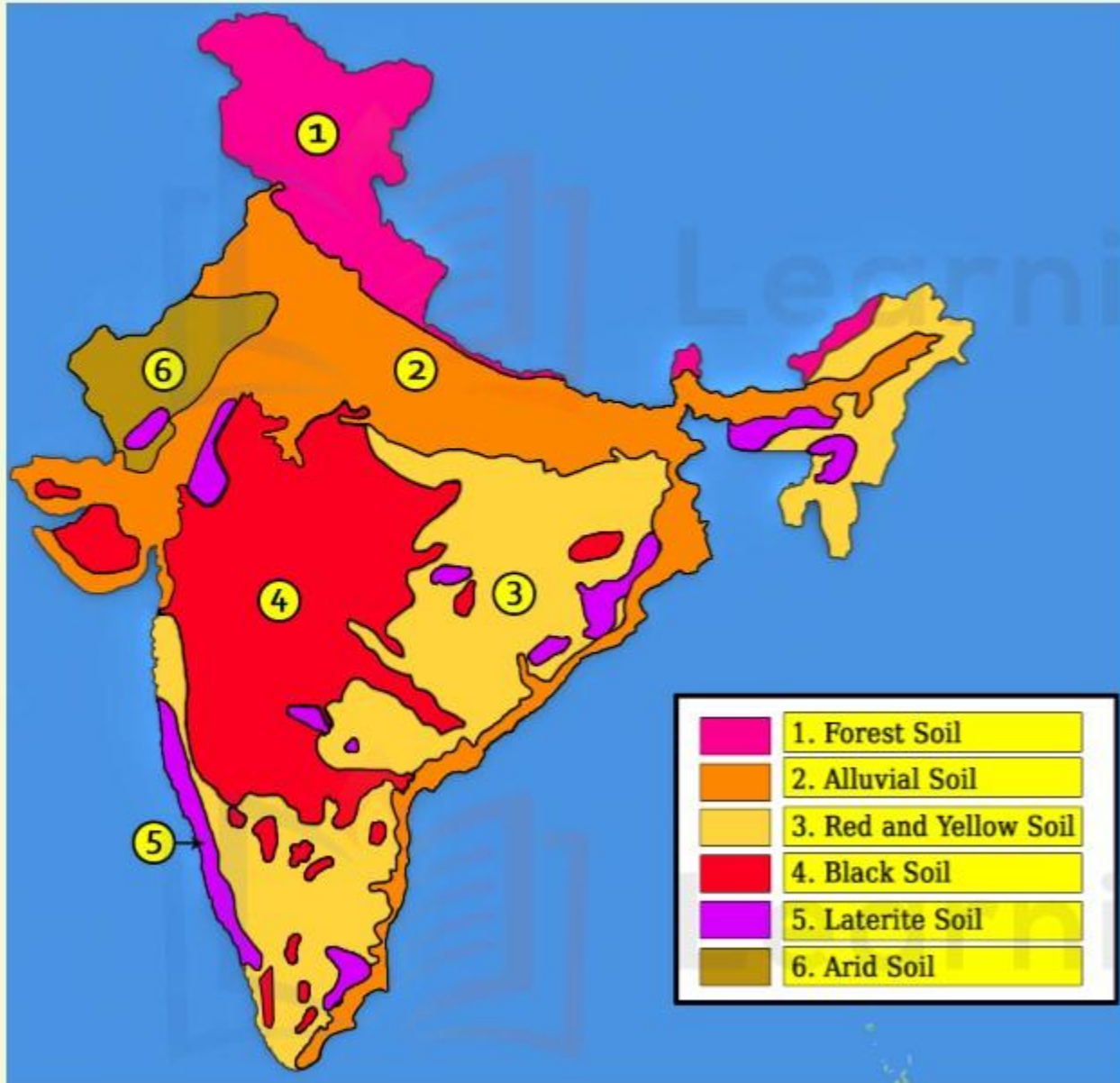
- (i) Alluvial soils
- (ii) Black soils
- (iii) Red and Yellow soils
- (iv) Laterite soils
- (v) Arid soils
- (vi) Saline soils
- (vii) Peaty soils
- (viii) Forest soils.

MAJOR SOIL TYPES OF INDIA



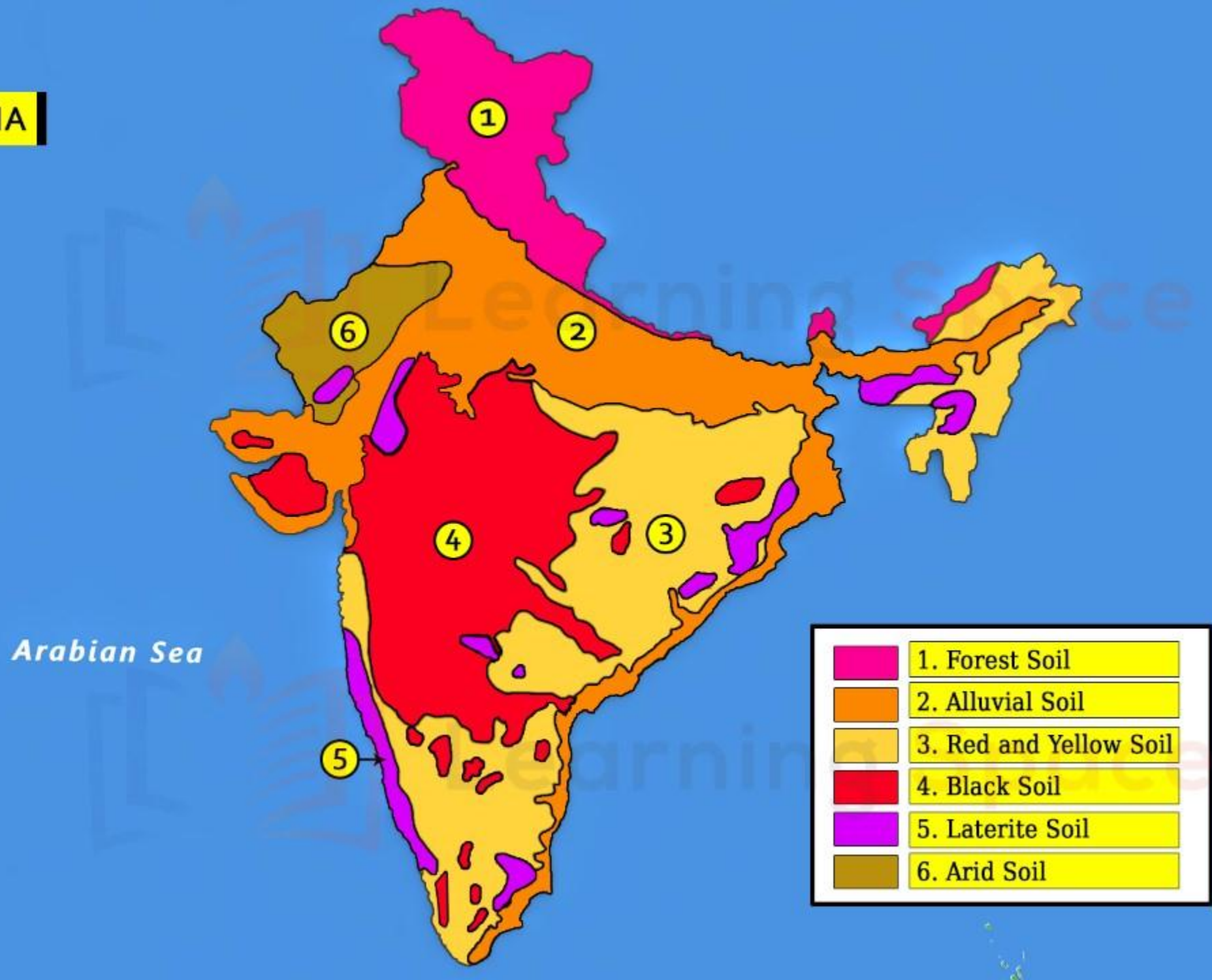
- 1** Geologically Indian Soil can be broadly divided into **Soils of Peninsular India** and **Soils of Extra Peninsular India**.
- 2** The soils of peninsular India are formed by **decomposition of rocks in situ** i.e. **directly from the underlying rocks**.
- 3** The soils of Extra Peninsular region are formed **due to the deposition by rivers and wind**. They are known as **transported or azonal soils**.

MAJOR SOIL TYPES OF INDIA



- 4** The major soil groups of India are
1. Alluvial Soils
 2. Black Soils
 3. Red Soil
 4. Lateritic Soil
 5. Forest and Mountain Soil
 6. Arid and Desert Soil
 7. Saline and Alkaline Soil
 8. Peaty and Marshy Soils.

MAJOR SOILS OF INDIA



MAJOR SOIL TYPES OF INDIA

ALLUVIAL SOILS



- 1 Alluvial soils are formed mainly due to the silt deposited by Indo-Gangetic-Brahmaputra rivers.
- 2 In coastal regions some alluvial deposits are formed due to wave action.
- 3 Rocks of the Himalayas form the parent materials. Thus these soils of transported origin.
- 4 They are largest soil group in India covering about 15 lakh sq km or about 46% of the total land area.
- 5 They support more than 40 % of India's population by providing the most productive agricultural lands.

MAJOR SOIL TYPES OF INDIA

ALLUVIAL SOILS

CHARACTERISTIC OF ALLUVIAL SOILS



- 1 The **parent material** of the alluvial soil is of **transported origin**.
- 2 The streams and waves bring with them the products of weathering of rocks from mountains and deposit them in the low-lying areas.
- 3 They are **immature and have weak profiles** due to their recent origin. Most of the soil is clayey and sandy soils.

MAJOR SOIL TYPES OF INDIA

ALLUVIAL SOILS

CHARACTERISTIC OF ALLUVIAL SOILS



KANKAR BEDS

- 4 Pebbly and gravelly soils are rare in this group.
- 5 A few occasional kankar beds are also present. Kankar beds are present in some regions along the river terraces.
- 6 Kankar are calcareous or calcium deposits found in the river beds.

MAJOR SOIL TYPES OF INDIA

ALLUVIAL SOILS

CHARACTERISTIC OF ALLUVIAL SOILS

7 The soil is porous because of its loamy nature. Porosity and texture provides drainage and other conditions favorable for bumper crops.

8 These soils are constantly replenished by the recurrent floods and support uninterrupted crop growth.

9 The chemical properties of alluvial soils makes the soil as one of the most fertile in the world.

10 The proportion of nitrogen is generally low but the proportion of potash & phosphoric acid are adequate. The proportion of iron oxide and lime vary within wide range.

MAJOR SOIL TYPES OF INDIA

ALLUVIAL SOILS

DISTRIBUTION OF ALLUVIAL SOILS



- 1 They occur all along the Indo-Gangetic-Brahmaputra plains.
- 2 In deltas of Mahanadi, Godavari, Krishna and Cauvery where they are known as Deltaic Alluvium.
- 3 Some Alluvial soils are found in the Narmada, Tapi and Northern Parts of Gujarat also.

Alluvial Soil



MAJOR SOIL TYPES OF INDIA

ALLUVIAL SOILS

CROPS IN ALLUVIAL SOILS



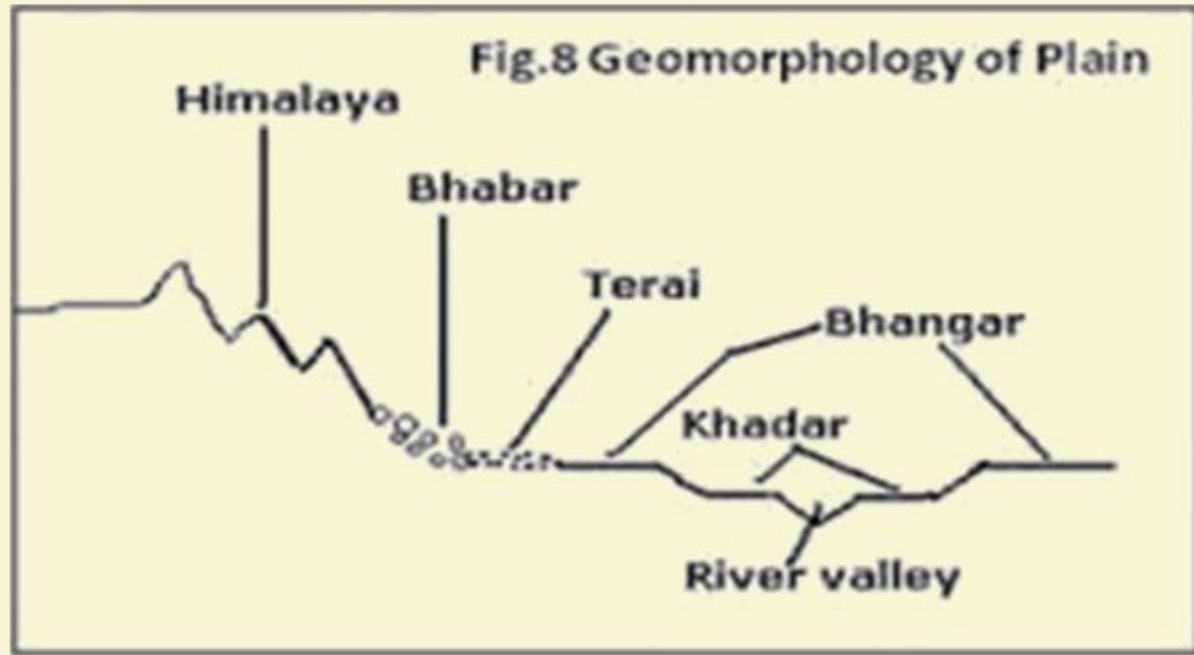
- 1 These soils are best suited for agriculture and respond well to canal and tube well irrigation.
- 2 They yield splendid crops of rice, wheat, sugarcane, cotton, jute, vegetables, oilseeds and fruits.

MAJOR SOIL TYPES OF INDIA

GEOLOGICAL DIVISION OF ALLUVIAL SOILS

Geologically the alluvium of great plains of India is divided into newer or **younger khadar** and older **Bhangar** soils.

BHABAR



- 1 The Bhabar belt is about **8 to 16 km** wide running along the **Shiwalik foothills**. It is a porous northern most stretch of soils in the Indo-Gangetic Plain.
- 2 Rivers descending from Himalayas deposit their load along the foothills in the form of **alluvial fans** which forms the Bhabar belt.

MAJOR SOIL TYPES OF INDIA

GEOLOGICAL DIVISION OF ALLUVIAL SOILS

BHABAR



BHABAR BELT

- 3 The porosity of the Bhabar is the most unique feature. The porosity is due to the deposition of huge number of pebbles and rock debris across the alluvial fans.
- 4 The *river stream disappear once they reach the Bhabar region because of this porosity.* Therefore the area is marked as the *dry river coasts except in the rainy season.*
- 5 The area is not suitable for agriculture only big trees with large roots thrive in this belt.

MAJOR SOIL TYPES OF INDIA

GEOLOGICAL DIVISION OF ALLUVIAL SOILS

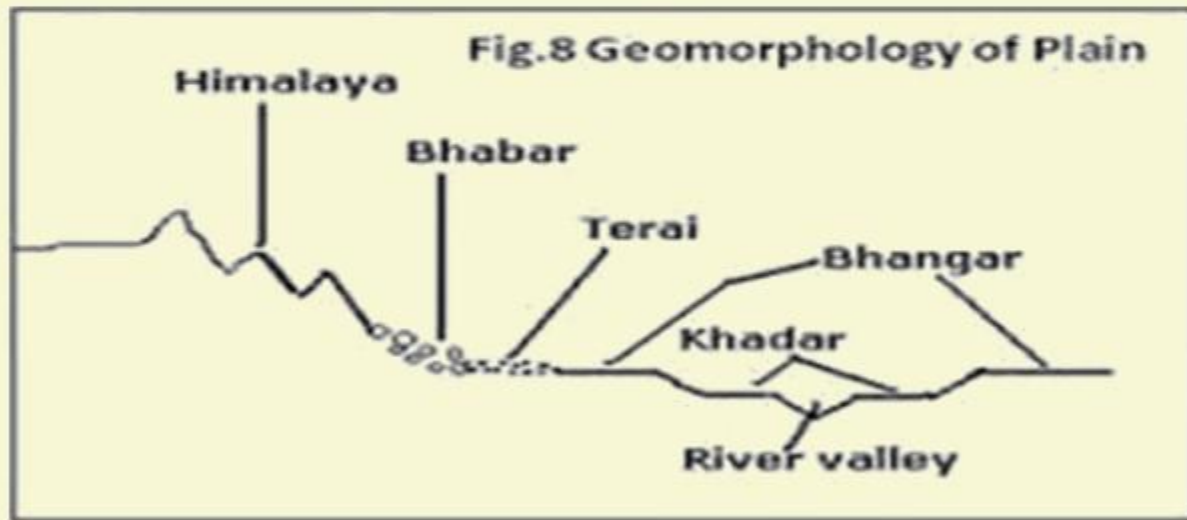
TERAI

- 1** Terai is a ill-drained,marshy and thickly forested narrow tract running south of the Bhabar belt.
- 2** It is around 15 to 30 km wide, the underground streams which disappeared in the Bhabar belt, reemerge in this belt and therefore this belt is swampy.
- 3** The terai are rich in nitrogen and organic matter but are deficient in phosphate.
- 4** These soils are generally covered by tall grasses and forests, but are suitable for number of crops such as wheat, sugarcane, rice, jute etc.
- 5** This thickly forested region provides shelter to a variety of wildlife.

MAJOR SOIL TYPES OF INDIA

GEOLOGICAL DIVISION OF ALLUVIAL SOILS

BHANGAR

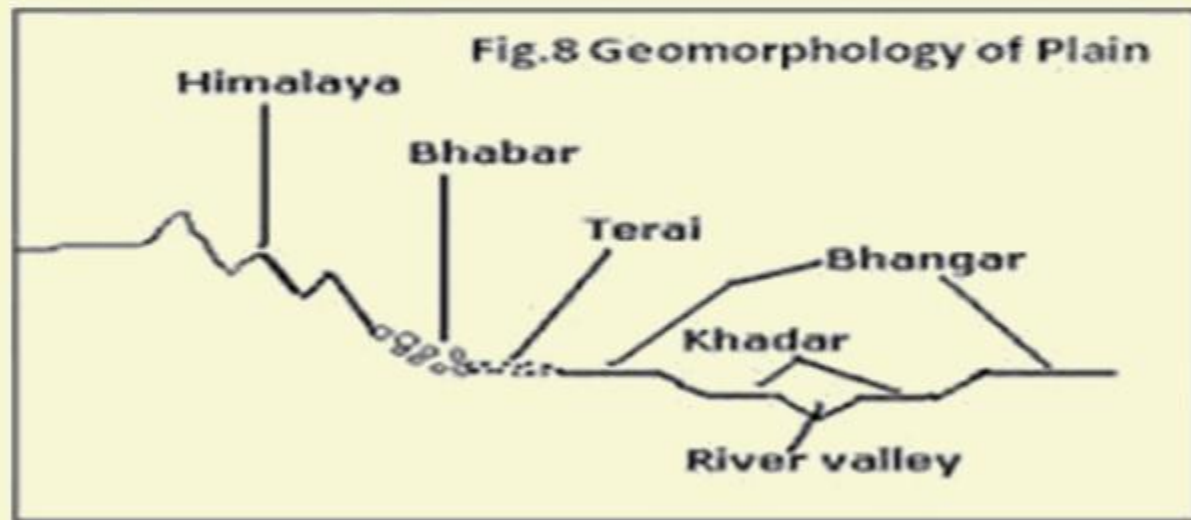


- 1 The **Bhangar** is **older alluvium** along the river beds forming the terraces higher than the flood plains above 30 mts more than the flood level.
- 2 It is of **more clayey composition** and is **generally dark colored**.
- 3 A few metres below the level of Bhangar are **beds of lime nodules** or calcium concretions which is known as the **Kankar**.

MAJOR SOIL TYPES OF INDIA

GEOLOGICAL DIVISION OF ALLUVIAL SOILS

KHADAR



- 1 Khadar is composed of newer alluvium and forms the flood plains along the river banks.
- 2 The banks are flooded almost every year, thus a new layer of alluvium is deposited with every flood. This makes the soils most fertile soils of the Ganges.
- 3 They are loamy soils more dry and less calcareous that is less Kankari.
- 4 A new layer of alluvium is deposited by the river floods almost every year.

MAJOR SOIL TYPES OF INDIA

DIFFERENCE BETWEEN KHADAR & BHANGAR

KHADAR (NEW)	BHANGAR (OLD)
1. The area in northern plains that lies below flood levels is known as Khadar which is new alluvium soil. Khadar is younger alluvium as flood waters deposit new layers every year.	1. Largest part of Northern Plains in India is made up of deposits of old alluvium soils known as Bhangar.
2. These soils are finer in texture	2. These are coarser in texture
3. Khadar is very fertile and ideal for intensive cultivation	3. These soils are less fertile as these are found in the higher reaches, about 30m above the flood level and present a terrace like structure.
4. Khadar is less calcareous/ carbonaceous i.e. they are less kankary.	4. Bhangar is full of kankar and calcareous structures
5. They are pale brown in colour and composed mostly of loams & sandy clays.	5. They are generally dark in colour and of a more clayey composition.

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For Suggestions:

suggestions@learningspace.in

To Contact us:

info@learningspace.in

OUR TEAM

G. V. Rao

K. Srikanth

D. Sunil Kumar

L. V. Krishna

Y. Deepthi

Amrita Naidu

M. Binukrishna

G. Ravi Babu

D. Joji

K. Victor Babu

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